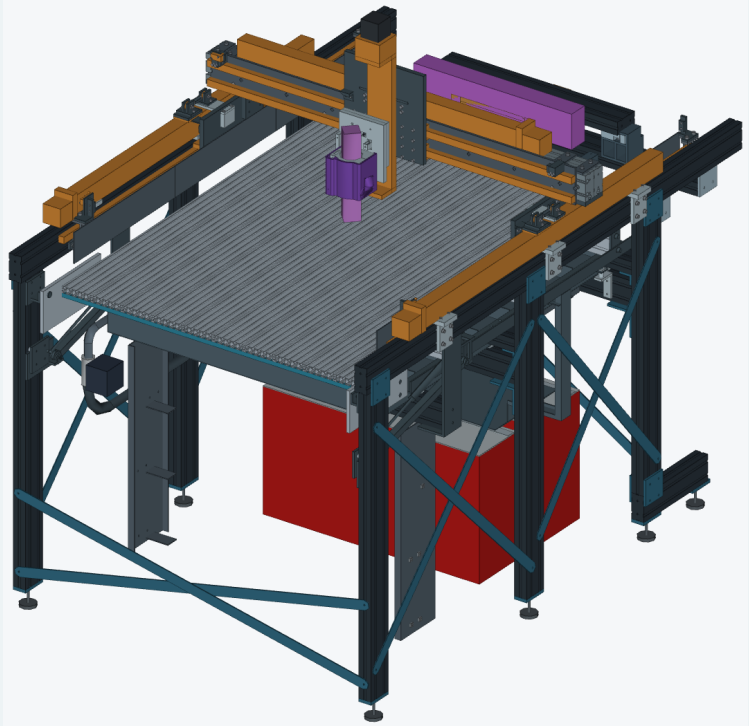
Integrated 4080 small table

Original owner head | Full HDPE / 20100 bed | Retractable ATC | 800 X / 1000 Y / 250 working Z

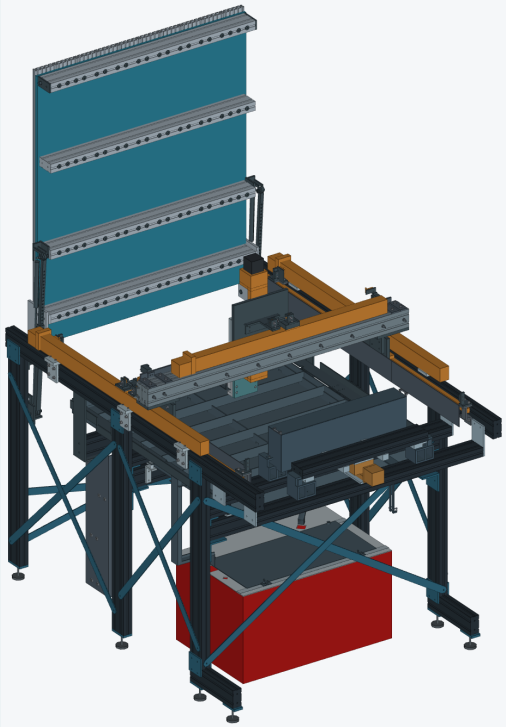
CURRENT ENGINEERING REVIEW - Supplier, control and delivered-budget gates remain.

WOOD / CARVING



Ten finished tracks, one HDPE layer, upright crossbars and removable billet head.

PLASMA / BED UPRIGHT



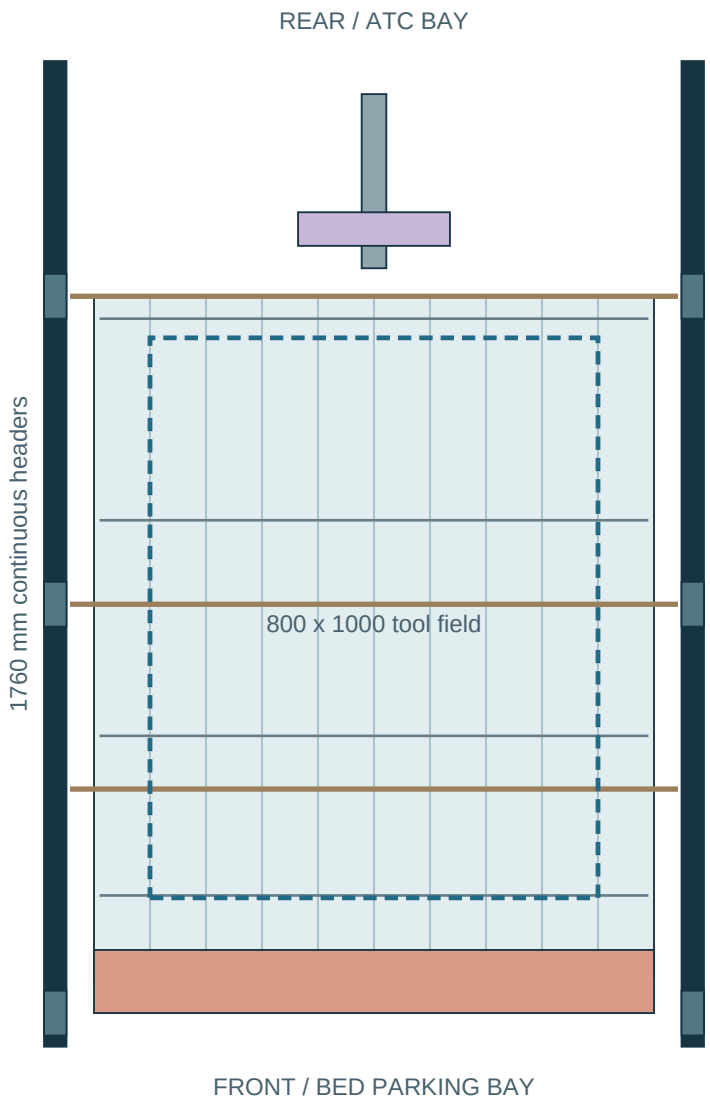
Bed parks in the front bay; covered magazine parks in the rear bay.

Rendered from the exported solid assemblies, including the original manufacturer magazine housing. Purple tool/spindle volumes are capacity context. The original owner boot remains at an unregistered reference pose.

The current source replaces earlier cassette beds and generic heads. It does not qualify the separate 4x 8 design or steel machining.

Layout and load paths

Permanent front parking bay and rear ATC bay keep the bed and guard changes within the machine plan.



Defined geometry

Bed: ten finished 1220 mm tracks; one 1000 x 1200 x 12.7 mm HDPE layer. Four 980 mm 4080 crossbars stand 80Z x 40Y at Y110/395/780/1140. Surface Z967.8.

Hinge: Y-100, Z967.8. Upright track/crossbar envelope Y-100 to 12.7; highest point approximately Z2247.8. Nominal deck mass 49.25-53.64 kg.

Frame: six posts and paired 1250 mm 4080 beams; continuous 1200 x 112 x 6.35 mm steel rail face, fitted lands and four upper ties. Actual rail pattern is transfer-fit.

Head: unchanged original clamp/boot; removable billet dock and owned-stock rib. Ideal local bending 0.093 mm at 100 N; joint slip and torsion are unqualified.

ATC: 200 mm slide; tool rows Y1100 /1300. Original manufacturer housing and pocket XY axes; slide-driven passive capture docks replace manual endpoint fasteners.

Front stops: cutting contact geometry is unchanged. Complete modules park 400 mm rearward, with guides stored and head removed, before bed tilt.

Analysis limits

The 100 N bed component screen is 0.156 mm versus 0.296 mm previously, using assumed I 536000 mm^4; at 50% retained I it is 0.203 mm. No HDPE composite action or full-machine stiffness is credited.

Pan/slats/drain are 26 mm lower; slat top Z852. The ideal full-brim pan screen is 1.156 mm. Actual sections, drilled ligaments, welds, joint slip, thermal effects and safe handling remain unqualified.

Each of two gas springs is a 150 lb /667 N candidate: compressed eye length at most 445 mm, expanded at least 700 mm. Received force curves, Z retention and interlocks require qualification.

Cost and release status

5 November 2026 source observations. Owned equipment excluded; shared tracks counted once; unknown charges remain unknown.

Currently sourced required goods	USD, before unknown charges
Two actual four-profile 1500 mm packs; cut eight to 1220	$259.98
One actual two-profile 1220 mm pack; retain both lengths	$82.99
US Plastic 42707: one 48 x 48 x 1/2-inch HDPE sheet	$191.86
RapidChange four-pocket ER11 Basic magazine	$400.00
Core goods	**$934.83**

Core plus already-staged hardware: $1046.91. Adding the required $50 tool setter gives $1096.91, before any 4080 frame, pan, remaining parts, freight or tax. The $1000 delivered cap fails. Core goods alone save $90.87 against the prior selected-source case.

Partial carts and delivery

16 variants /33 packs /538 pieces: $455.05 goods, including the same $342.97 tracks. AliExpress: 13 variants, $78.32. Amazon: 3 variants, $376.73. No order placed; remaining BOM items are unstaged.

Long 1500 mm stock: **December 9-24** final cart estimate, superseding November. Selected Amazon lines show free Prime to 30510. Ali exact Alto address is unconfirmed. HDPE/ATC freight and all tax are unknown.

Owned cost exclusions

Owned motion, guides, drives/controller, spindle/VFD, working head, enclosure, cutter/torch, compressor and selected slide are excluded. Two aluminum squares have 14 exclusive blank allocations. Bed materials and ATC have no ownership credit.

Physical and operating gates

1. Register boot home, spindle nose and actual ER11 nut/socket engagement. Confirm received kit revision, state sensing, rail and slide patterns.

2. Qualify actual extrusion/drilled sections, slot nuts, welds, joint slip, full rigidity, power-loss Z retention and controlled bed handling.

3. Verify receiver water/temperature/overflow, two 150 lb-each struts, valve/fittings, module shielding, VFD reverse/stop, isolated plasma control and interlocks.

4. Finish the delivered quotation and remaining selections. Valid exported solids and nominal clearance do not establish fabrication or operating release.

Current source, checks, stock and status: public project / variants / integrated-4080-small